

Jovan Cvijanović

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Education

Faculty of Technical Sciences, University of Novi Sad – Software Engineering

2023 – 2027 (expected)

Languages: English (Cambridge C2 Proficiency), Serbian (native)

Projects

cyclops – Monocular Depth Estimation Study | *Python, PyTorch, Stable Diffusion, I-JEPA, DepthAnything V2* **GitHub**

- Ran a **comparative study of monocular depth estimation** on NYU Depth V2 across five approaches: a supervised ResNet-50 baseline, frozen **Stable Diffusion** and **I-JEPA** features, their fusion, and DepthAnything V2 as a SOTA reference.
- Evaluated **robustness under degraded inputs** (fog, blur, under-exposure) to measure how each method holds up outside clean conditions.
- Built a **config-driven experiment pipeline** with W&B tracking in a 2-person team.

RAG Systems – Research to Deployment (CodeRAG & Otto RAG) | *Python, LangChain, PostgreSQL, pgvector,*

Ollama, FastAPI, OAuth2

CodeRAG | Otto RAG

- **Research (CodeRAG):** built a RAG QA system for codebases and improved **Recall@10 from 63.7% to 88.8%** through LLM-generated code summaries, enriched metadata, and embedding/chunking experiments.
- **Deployment (Otto RAG):** productionized those learnings into a **deployed personal RAG** over Gmail and Google Drive with **PostgreSQL + pgvector**, local inference via Ollama, per-user data isolation, and an **OAuth2-authenticated FastAPI** service.
- Implemented a shared **agentic query pipeline** with query rewriting, document relevance grading, hallucination checking, and out-of-scope handling.

Interia – AI-Powered Room Redesign | *React, TypeScript, Three.js, Sentence Transformers, FastAPI, Vercel*

GitHub

- Built the **React frontend** with interactive 3D point cloud visualization (Three.js), video upload, and real-time furniture replacement previews.
- Developed a **furniture data pipeline** over a 45,000+ product catalog and implemented **semantic search (RAG)** using sentence transformers for intelligent product matching.
- Managed **deployment infrastructure** on Vercel with Docker-based GPU backend.

word2vec – CBOW & Skip-Gram with Negative Sampling | *Python, NumPy*

GitHub

- Implemented **CBOW and SGNS from scratch in pure NumPy** — no ML frameworks — with full training loop, negative sampling, sub-sampling of frequent words, and learning rate decay.
- Evaluated on **Google analogy** (19k questions) and **WordSim-353**; best results: **44.2% analogy accuracy** (CBOW) and **Spearman $\rho = 0.731$** (SGNS) on text8.
- Built an interactive **REPL for nearest-neighbour queries and vector arithmetic**, a **W&B sweep** for hyperparameter search, pretrained embedding releases, and a **pytest test suite**.

gviz – Graph Visualizer | *Python, Django, D3.js, Plugin Architecture*

GitHub

- Architected the **core platform and abstract plugin API** in a 4-person team, defining the interfaces all plugins implement and building the plugin management, workspace, and graph filtering systems.
- Developed the **first data source plugin** (JSON format with cyclic structure support) and designed the **complete UI layout** in Django that teammates extended with additional functionality.

Lavugio – Ride-Sharing Platform | *Java, Spring Boot, Angular, PostgreSQL, WebSocket, Leaflet, JWT*

GitHub

- Developed **backend services and repositories** in Spring Boot for a full-stack Uber-like application built in a 3-person team.
- Built **frontend features** in Angular including real-time WebSocket chat, push notifications, and live map tracking with Leaflet.
- Developed the **Android companion app** in Java with Material Design 3.
- Wrote **unit and E2E tests** with JUnit, Jasmine, and Selenium.

Experience

- Microsoft – Applied ScientistBelgrade, Serbia (Hybrid) | July 2026 – Present
- Member of the **Applied Science team**, working on applied AI research and its deployment into production systems.
- Freelance C++ Tutor
- Conducted 1-on-1 lessons in C++ covering core language features, memory management, and object-oriented programming.

Non-Industry Relevant Experience

- Mediterranean ML Summer School 2025 – ParticipantSplit, 08–12 September 2025
- Selected as one of **300 participants** out of ~1700 applicants.
- Competitive Programming Task Author2024 – Present
- Designing algorithmic problems and organizing competitive programming competitions in Bosnia and Herzegovina.
 - Built **Python tooling** for automating task generation, competition management, and report generation.
 - Co-authoring a **training syllabus** covering algorithms and data structures for national-level competitors.
- Link

Awards

- Garaža Frontier Hackathon (Belgrade, Serbia) – 1st place, \$10,000 prizeDecember 2025
- Won with **Interia**, an AI-powered room redesign app; 48-hour hackathon with 30+ competing teams.
- National Olympiad in Informatics (Republic of Srpska) – 1st place2022
- Mathematical Grammar School Cup (Belgrade) – Bronze Medal in programming2020